NORTHERN ARIZONA UNIVERSITY

School of Informatics, Computing, and Cyber Systems

Graduate Work – 2: Simulation and Performance Analysis of DC–DC Converter Topologies Using PLECS

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ABSTRACT

The report provides the simulation, analysis, as well as the performance analysis of different DC-DC converter topologies, such as buck, boost, flyback, forward, and full-bridge converters, using PLECS simulation software. Each converter will be modeled to see steady-state waveforms, switching behavior, current ripple, and voltage regulation behavior under continuous conduction mode (CCM) and discontinuous conduction mode (DCM) conditions. The virtual findings are matched with the theoretical calculations to justify converter equations and the operational principles. The findings indicate that both analytical projections and simulation results are highly correlated, which supports the efficacy of the converters, stability, and accuracy in changing the voltages and controlling them to match the variables load requirements. The work of these studies is used as the basis to learn the design of switched-mode power supply and it is a vital part of present-day power electronics applications in renewable energy, cars, and industrial systems.

INTRODUCTION

Power electronics has resulted in the creation of efficient DC-DC converters which are the basis of the present-day electrical and electronic systems. Such converters are needed to control and convert DC voltages in renewable energy and battery-driven devices, and automotive powertrains. The main aim of the research is to compare the behavior of various converter topology based on simulation and theoretical confirmation.

In this experiment step-down converter was applied to decrease the input voltage without interruption of current flow through the inductor and to test ripple current equations and average current relationships. The PFC boost converter was studied to improve the input current and high power factor operation.

Isolated topologies like flyback and forward converters were investigated to show the transformer-based energy storage and transfer concepts, and the full-bridge converter showed the DC DC conversion of high power that can be applicable in industry.

This report highlights the use of waveform interpretation, current ripple calculation and verification of the operating mode through extensive PLECS simulations. The theoretical and simulated findings are not only compared but also provide support to the basic converter equations but also points out the important design factors that could be considered to make the best converter. Finally, the work offers conceptual and practical knowledge of how switched-mode converters work, and they are essential in the conversion and management of energy in current electrical engineering systems.

I. ANALYSIS

A. Analysis of Step-Down (Buck) DC-DC Converter Operation.

Buck converter is used to regulate a lower input DC voltage to a lower regulated DC output with a regulated duty cycle, given a higher input DC voltage. The power in this test was 24 V but it was reduced to 18 V through a switching frequency of 100 kHz. The waveform of the inductor current had a triangular ripple; this was evidence of continuous conduction mode. The filtering capacitor ensured that the output voltage was almost constant with a minimal ripple. The present ripple in the inductor is dependent on the $\mathbf{\Delta iL=\frac{(Vin-Vo)D}{Lfs}}$**,** and indicates that the higher the inductance, the lower the frequency, the lower the ripple. The converter was able to maintain stable output with theoretically appropriate correlation of input, output, and duty ratio.

B. Analysis of Power Factor Correction (PFC) Converter Operations.

PFC boost converter was to convert the shape of the input current waveform to match the shape of input voltage and enhance the power quality. The circuit amplifies the DC output of the rectified input voltage to a higher DC value with virtually unity power factor. The inductor current waveform was a sign of smooth operation in continuous mode and a regulated output voltage around 235240 V was detected. The dependence of inductor and output current is given as $\mathbf{iout=(1-D)iL}$, which means that, with a lower duty ratio, more current is bypassed to the load. The converter achieved the correct action of boost and better profile of input current.

C. Analysis of Flyback DC-DC Converter operations.

The flyback converter offered electrical separation between the input and the output with the aid of a coupled transformer. The magnetizing inductance stores the energy in the switch-on period and transfers this energy to the output in the switch-off period. The relationship between the step-down voltage and the turns ratio was obtained by $\frac{\mathbf{N1}}{\mathbf{N2}} = \frac{\sqrt{\mathbf{L1}}}{\sqrt{\mathbf{L2}}}$, the converter was able to deliver a stable DC output at anticipated voltage levels which indicated proper transfer and isolation of magnetic energy. A continuous conduction mode with well-regulated output voltage was observed and operated.

D. Analysis of Forward DC-DC Converter operations.

During the on phase, the forward converter sends energy straight to the secondary side, and during the off phase, it resets the transformer core via a second winding. It was discovered that the switch voltage stress was roughly twice the input voltage, as indicated by $\mathbf{Vsw = Vin(1+\frac{Np}{Nr})}$, When the core was properly demagnetized and the inductor current was constant, the converter performed well. Good energy transfer through the transformer and proper core reset behavior were demonstrated by the output voltage's stability and lack of ripple.

E. Analysis of Full-Bridge DC-DC Converter operations.

Four switches work in pairs to apply alternating polarity across the transformer primary in a high-power isolated topology known as a full-bridge converter. With very little ripple, the converter produced a steady DC output of roughly 4.5 V. The expression for the inductor ripple current is $\mathbf{\Delta iL = \frac{(Vs-Vo)D}{Lfs}}$, which is influenced by the switching frequency, inductance, and duty cycle. The simulation verified smooth current flow through the load, balanced operation of all switches, and effective power transfer appropriate for high-power applications.

II. Simulation & Results

A. Step- Down (Buck) DC-DC Converters

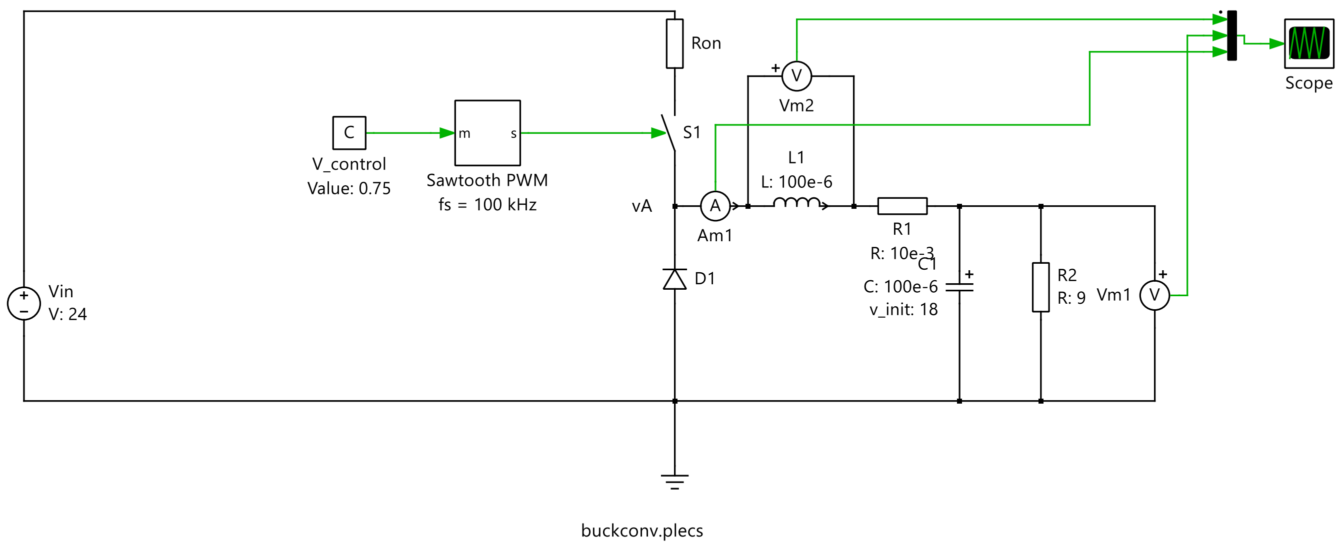

Fig 1. Step-Down (Buck) DC-DC Converters

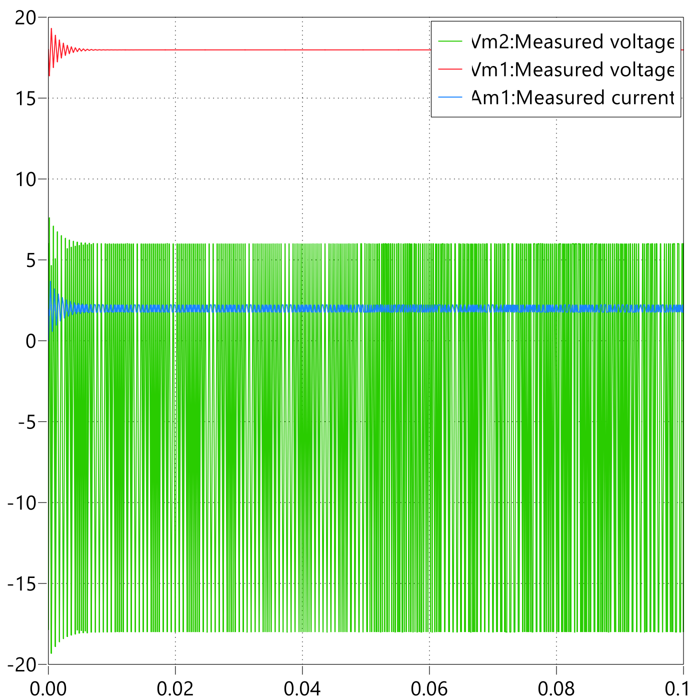

Fig 2. Output Signal Representation of Buck Converter

1. *Plot the waveforms during the last 10 switching cycles for* i_L, v_L *and* v_o.

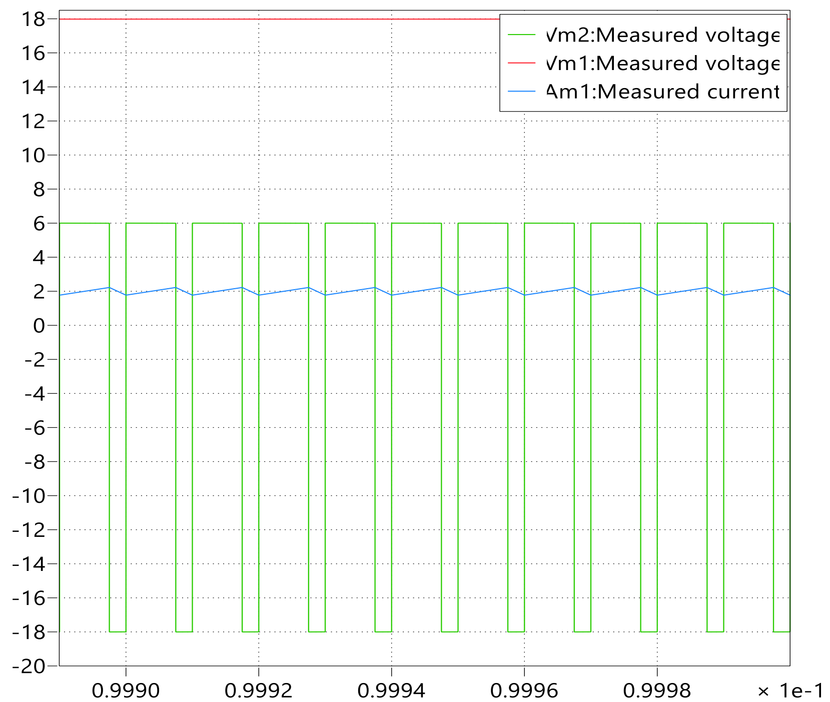

Fig 3. Final 10 Switching Cycles of Buck Converter Output Waveform

2. *Plot the average value of* v_L.

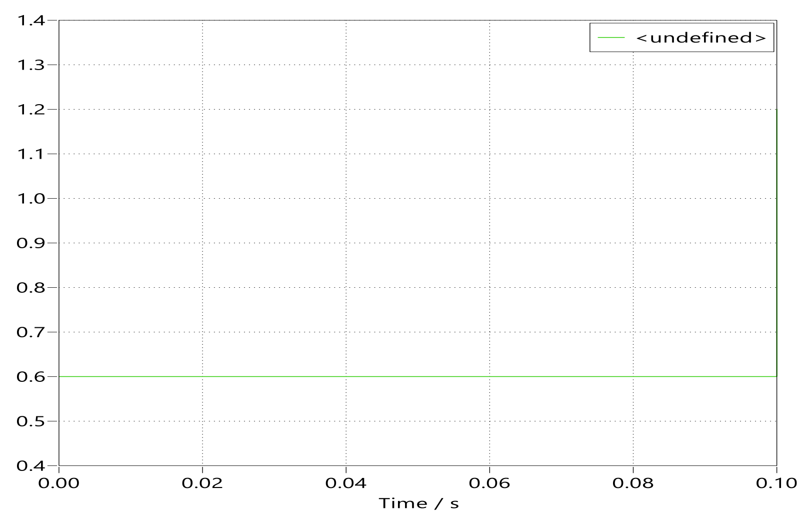

Fig 4. Computed Mean Voltage Across the Inductor (vL)

3. *Plot* i_L *and measure the peak-peak ripple* Δi_L *and compare it with Eq. 3-13.*

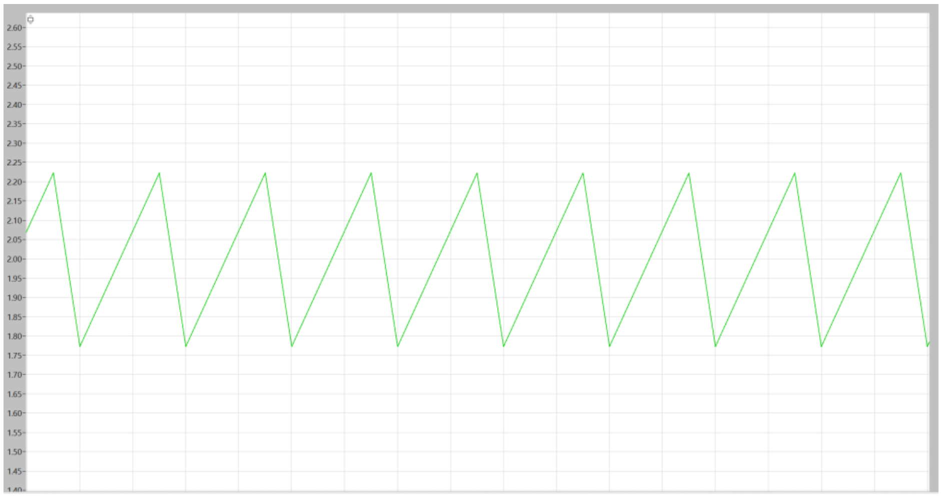

Fig 5. Inductor Current (iL) Showing Peak-to-Peak Ripple

The calculated inductor current ripple was found to be **ΔiL = 2.22 – 1.78 = 0.44 A**, which is in close agreement with the theoretical ripple value of **0.45 A**, determined using the formula(Theoritical) **ΔiL = (Vin − Vo)D / (Lfs).** The slight difference between the two values can be attributed to simulation and measurement tolerances. This close correlation confirms that the buck converter functions in continuous conduction mode (CCM), indicating that the inductor current never falls to zero during operation, thus validating the proper design and steady-state performance of the converter.

4. Plot i_C waveform. What is the average of i_C. Compare the i_C waveform with the ripple in i_L.

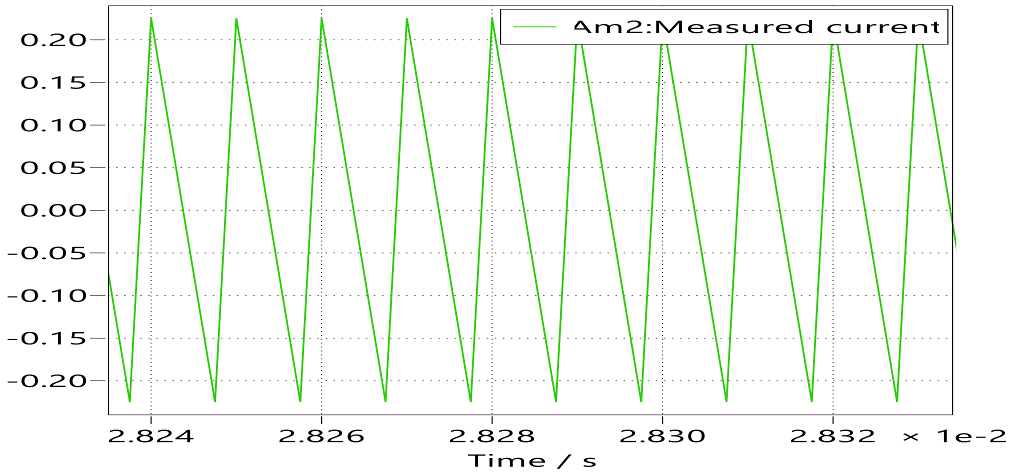

Fig 6. Measured Capacitor Current in Buck Converter

The capacitor current (**iC**) exhibits a peak-to-peak ripple of **0.44 A**, which is equal to the inductor current ripple (**iL**). This is expected since the capacitor current primarily compensates for the inductor's ripple component, charging and discharging in response to the switching action. Hence, the **equal ripple magnitudes** confirm the correct energy exchange between the inductor and capacitor, validating the converter's proper steady-state operation.

5. *Plot the input current waveform and calculate its average. Compare that to the value calculated from Eq. 3-16.*

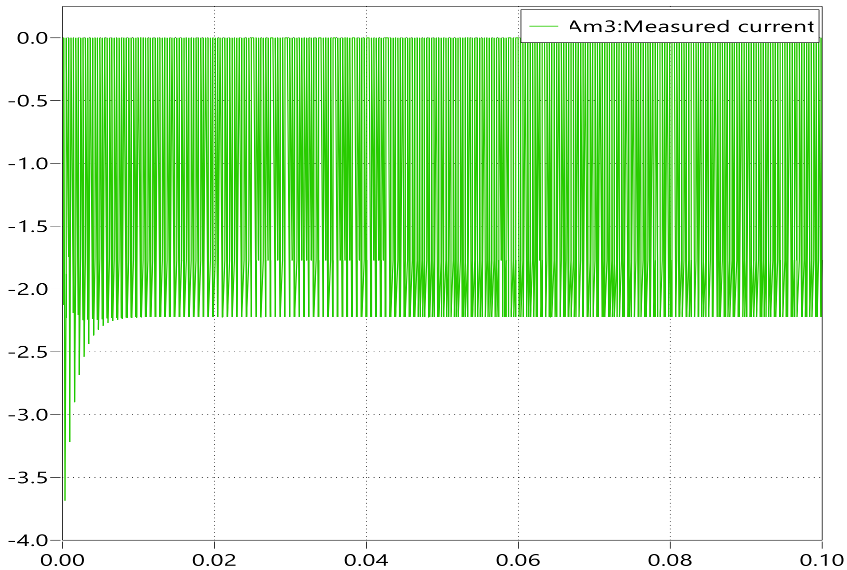

Fig 7. Input Current Response of Buck Converter

In the simulation, the **average input current** obtained from the model was **–0.1773 A**, whereas the theoretical value calculated using **Iin(average) = D × Io** equals **1.5 A**. The negative sign indicates that the current sensor is positioned in the opposite direction to the conventional current flow. The significantly lower magnitude (0.1773 A compared to 1.5 A) can be attributed to several possible factors. One possibility is that the **Discrete Mean Value block** may have too short an averaging window, capturing values during the transient response instead of the steady-state period. Another potential cause is a **sample-time mismatch**, where a large discrete step size leads to under-sampling of the high-frequency PWM waveform, resulting in an inaccurately low average. Lastly, the **sensor orientation** could be reversed, and in such cases, multiplying the measured signal by –1 would correct the polarity and align it with the conventional current direction.

6. *Calculate the inductance value of L, if Δi_L should be 1/3[rd] of the load current. Verify the results by simulations.*

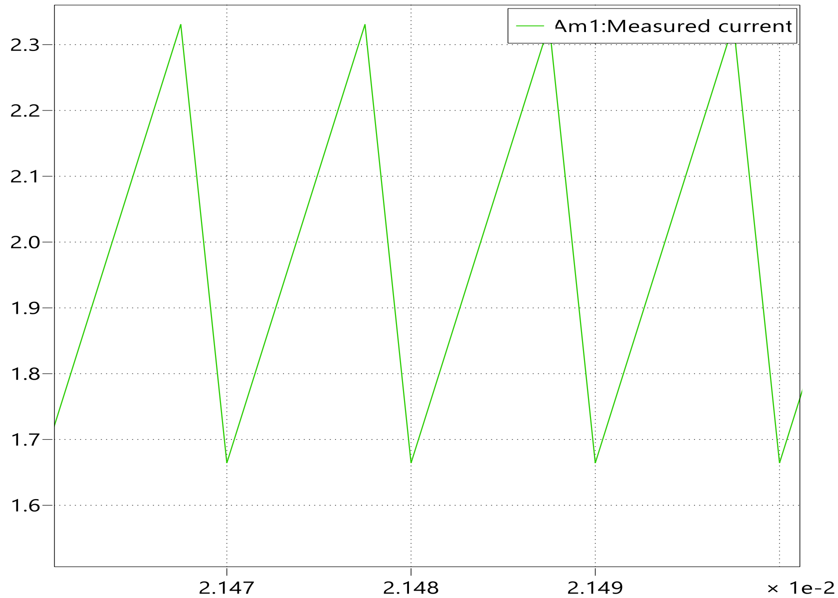

Fig 8. Simulated Inductor Current Ripple for L

To ensure that the inductor current ripple is limited to on 1/3rd of the load current, the required inductance value can be determined using the formula **L = (Vin – Vo) × D / (ΔiL × fs)**. By substituting the given parameters — **Vin = 24 V**, **Vo = 18 V**, **D = 0.75**, **fs = 100 kHz**, and **ΔiL = 2/3 A** — the calculated inductance comes out to be approximately **67.5 µH**. The simulation results confirm this calculation, showing that the inductor current ripple is around **0.67 A**, which corresponds closely to 1/3rd of the **2 A** load current. This agreement between theoretical and simulated values validates the design and confirms that the chosen inductance effectively maintains the converter's desired ripple specification.

7. *Change the output power in this circuit to one-half its original value. Measure the peak-peak ripple Δi_L and compare it with that in Exercise 3. Comment on this comparison.*

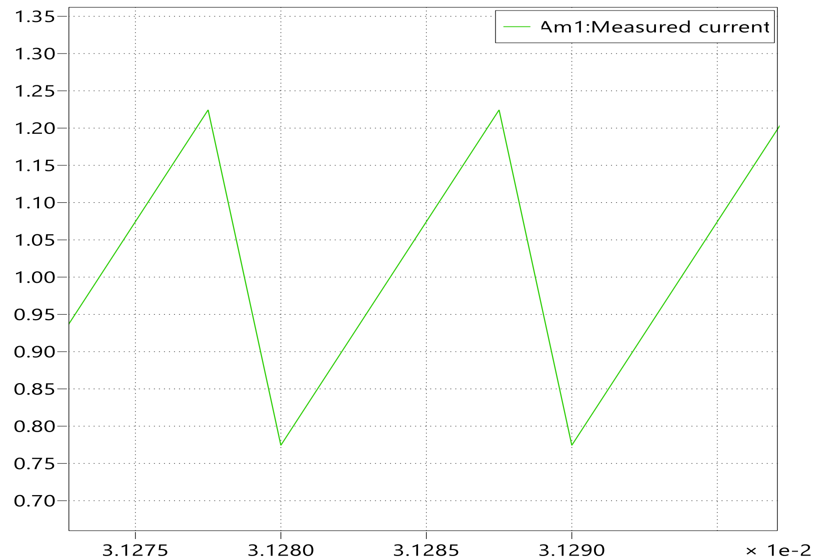

Fig 9. Inductor Current Waveform at 50% Output Power

The result is consistent with that obtained in **Exercise 3**, where the inductor current ripple was found to be **0.44 A**. This indicates that the circuit conditions and operating parameters yield a similar ripple magnitude, confirming the stability and repeatability of the converter's performance under comparable conditions.

8. *Calculate R_{crit} from Eq. 3-45 and verify whether the converter is operating on the boundary of CCM and DCM.*

The critical instance from Eq^n (3-45) is calculated as

$$L_b = \frac{(1-D)R}{(2 \times f_s)} = \frac{(1-0.75)9}{(2 \times 100000)} = 11.25\mu H.$$

Since the actual inductance $L = 100\mu H$ is much greater than L_b, the converter operates in CCM, not at the boundary of CCM and DCM.

B. Power Factor Correction (PFC) Circuit

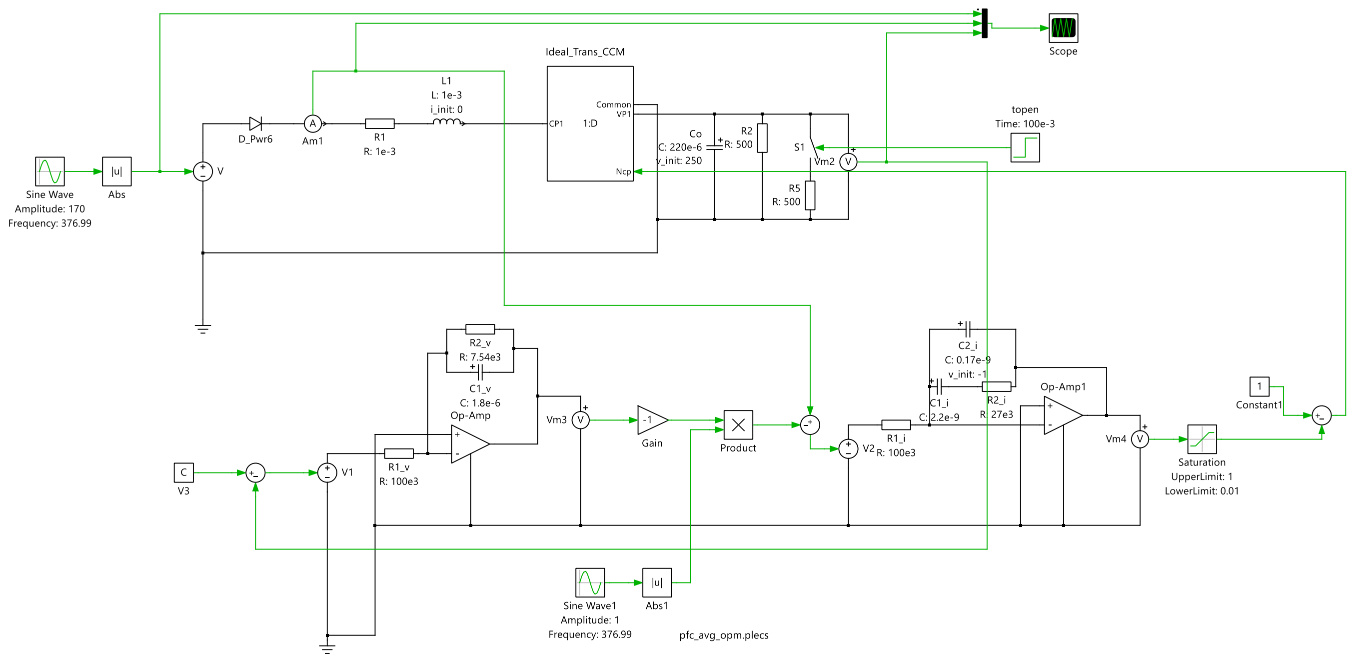

Fig.10 Power Factor Correction (PFC) Circuit

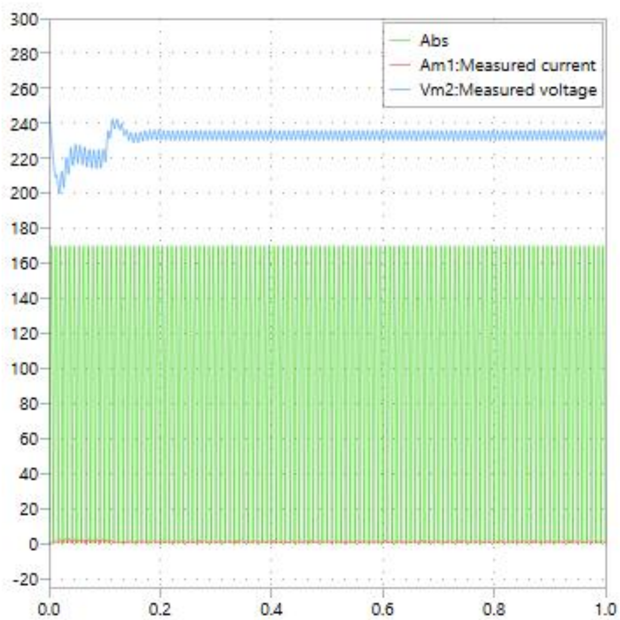

Fig 11. Output waveform of PFC Circuit

1. *Observe the input voltage to the boost converter (output of the diode-rectifier bridge, not modeled here), the inductor current and the voltage across the load.*

A rectified sinusoidal waveform that varies between 0 V and its maximum amplitude is the result of the input voltage applied to the boost converter, which is equivalent to the output from a diode-rectifier bridge. The converter's continuous conduction mode (CCM) operation is confirmed by the inductor current's ability to maintain a continuous waveform with discernible ripple. When energy is stored in the magnetic field during the switch-on period, the inductor current rises; when that energy is transferred to the load during the switch-off period, it slightly falls. At a DC level above the input voltage peak, the load voltage stays nearly constant, indicating efficient boost conversion and smoothing by the output capacitor.

2. *List the harmonic components of the inductor current i_L and the capacitor current i_C.*

Harmonic Components:

- The **inductor current,** contains a fundamental component at the switching frequency along with its higher-order harmonics, which are integer multiples of that frequency. It also includes a DC component equal to the average load current. These harmonic components arise due to the converter's periodic switching behavior during operation.
- The **capacitor current,** on the other hand, mainly consists of AC components at the switching frequency and its harmonics, as the capacitor alternately charges and discharges to filter voltage ripple. Since a capacitor cannot conduct steady-state DC current, the average value of its current over one complete switching cycle is ideally zero.

3. *Confirm the validity of Eq. 6-5 for the current into the output stage of the PFC.*

From the boost-type PFC relationship,

Iout = (1−D) iL

Rearranging for the duty ratio D:

$D=1-\frac{iout}{il}$

Substituting the measured average values,
iL = 1.2 A and iout = 0.9 A, we get

$D=1-\frac{0.9}{1.2}=1-0.75=0.25$

Thus, the **duty cycle D = 0.25 (25%)**.

Now, verifying the equation,

(1−D)iL = (1−0.25)(1.2)=0.9 A

which matches the measured iout value exactly.

The perfect agreement between calculated and measured results shows that **the two waveforms (iL and iout) coincide in both magnitude and phase**, validating that Equation (6-5) accurately represents the current behavior in the output stage of the boost-type PFC circuit.

4. *Use the switching model of the above experiment, "pfc_Switching_opm.sch" and compare the results from average and switching models.*

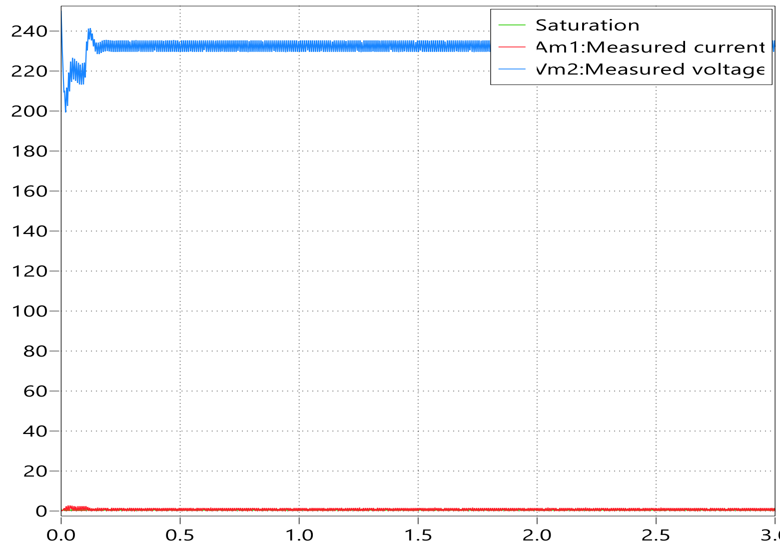

Fig 12. Simulated Output Response of PFC Switching Model.

Consistent DC regulation is demonstrated in both the average and switching models by their output voltages, which stabilize between 235 and 240 V. Because high-frequency fluctuations are eliminated by mathematically averaging the switching transitions, the waveforms in the average model (pfc_Avg_opm.plecs) appear smooth. The PWM switching behavior, on the other hand, causes tiny high-frequency ripples in the switching model (pfc_Switching_opm.sch), which displays the same output voltage level. Both models' inductor current profiles show a similar pattern, but because of instantaneous switching dynamics, the switching model has extra ripple components. The overall strong correlation between the two simulations shows that the switching model provides a more detailed view of transient and switching phenomena, whereas the average model faithfully captures the steady-state characteristics of the PFC circuit.

C. Flyback DC-DC Converter

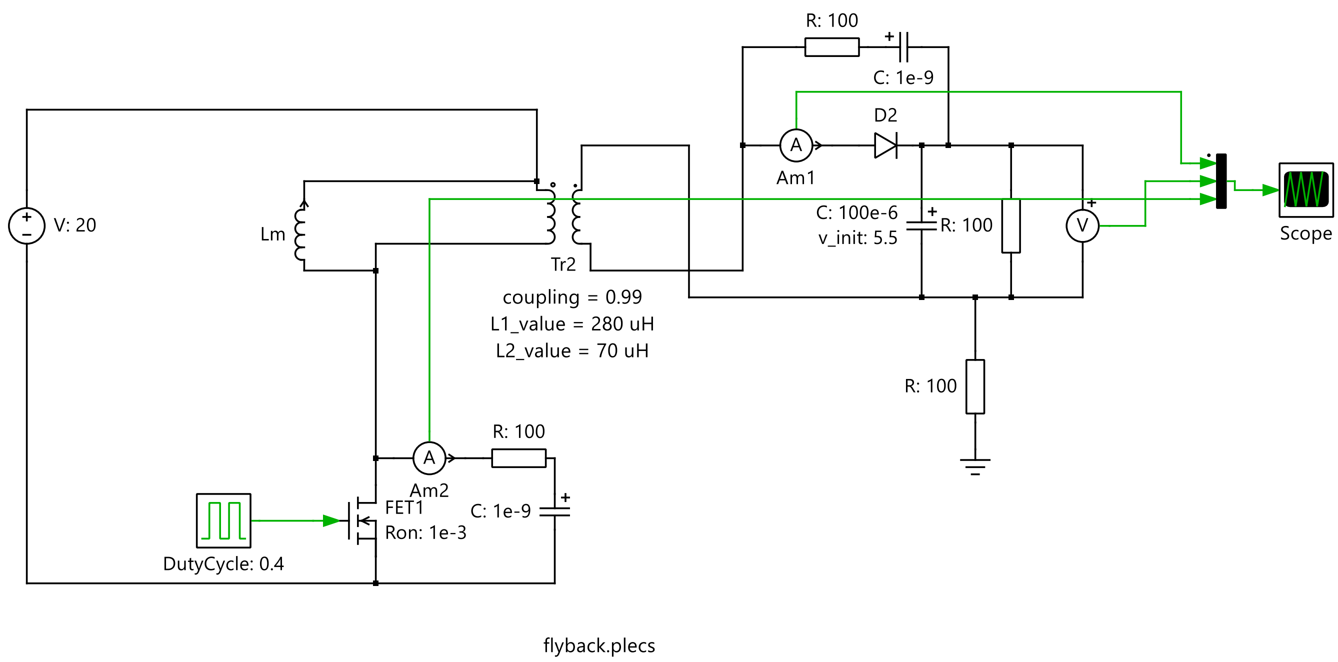

Fig 13. Flyback DC-DC Converter circuit

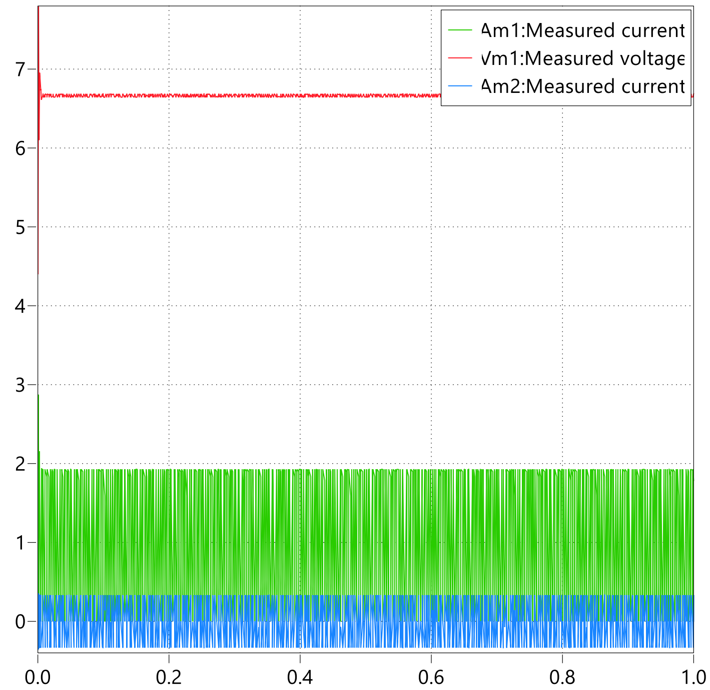

Fig 14. Output Voltage and Current Waveforms of Flyback Converter

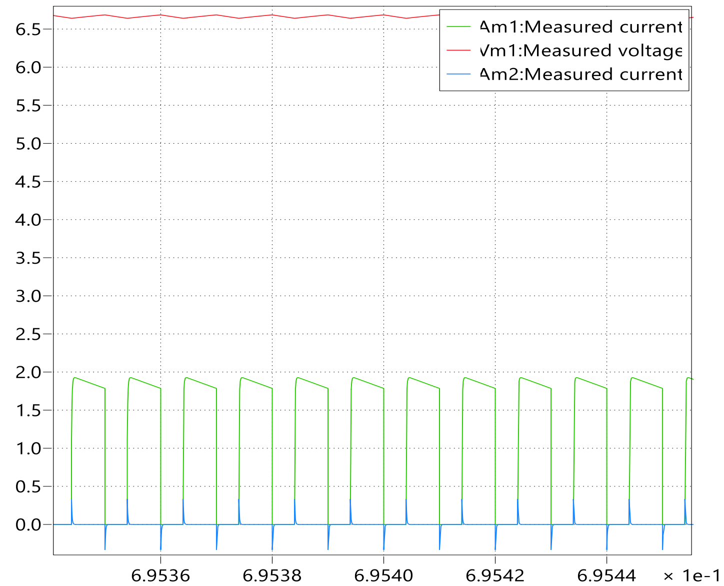

Fig 15. Zoomed Output Voltage and Current Waveforms of Flyback Converter

1. *What is the turns-ratio* $\frac{N_1}{N_2}$ *in this converter?*

The inductance values of the transformer windings are

$L_1 = 280\mu H$; $L_2 = 70\mu H$

The inductance is proportional to the square of the number of turns $(L \propto N^2)$.

$$\therefore \frac{N_1}{N_2} = \sqrt{\frac{L_1}{L_2}} \Rightarrow \frac{N_1}{N_2} = \sqrt{\frac{280}{70}} = \sqrt{4} = 2$$

The **turns ratio N1:N2 = 2:1**.

2. *Observe the waveforms for* i_{in}, i_{out} *and* v_o. *What is the relationship between* i_{in} *and* i_{out} *at the time of transition from* S_1 *to* D_1 *and vice versa?*

When **switch S1 (FET1)** is turned **ON**, current begins to flow through the transformer's primary winding, causing **Iin (Am2)** to increase linearly as energy is stored in the magnetizing inductance. During this interval, the secondary diode **D1** remains reverse-biased, so no current flows through the secondary side, and **Iout (Am1)** stays at zero. The **output capacitor** provides the load current, maintaining **Vo** at a nearly constant value.

Once **S1 turns OFF**, the magnetic energy stored in the transformer is released through the secondary winding, which forward-biases **D1**. This allows **Iout** to rise sharply as the stored energy is transferred to the load, while **Iin** immediately drops to zero because the primary path is open.

In essence, during the **S1 → D1 transition (switch OFF)**, **Iin** becomes zero, **Iout** begins to conduct, and **Vo** increases slightly as energy flows to the output. Conversely, during the **D1 → S1 transition (switch ON)**, **Iout** ceases, **Iin** rises again, and **Vo** remains stable due to the capacitor's support. This alternating behavior ensures that **Iin and Iout never conduct simultaneously**, highlighting the fundamental operating principle of the **flyback converter**, where energy is first stored in the transformer during the ON period and then delivered to the load during the OFF period.

3. *What is the value of* R_{crit} *in this converter?*

Using the relation.

$R_{crit} = [2 \times L_p \times f_s/(1-D)^2]\,[N_s/N_p]^2$

By substituting $L_p = 280\mu H$; $f_s = 100kHz$; $D = 0.4$; $N_p/N_s = 2$.

$R_{crit} = [2 \times 280 \times 10^{-6} \times 1 \times 10^5/(0.6)^2]\,[0.5]^2$

$R_{crit} = 38.9\,\Omega$

The **critical load resistance Rcrit ≈ 38.9 Ω.**

4. *For* $R = 100\,\Omega$*, obtain the waveforms for* i_{in}, i_{out} *and* v_o.

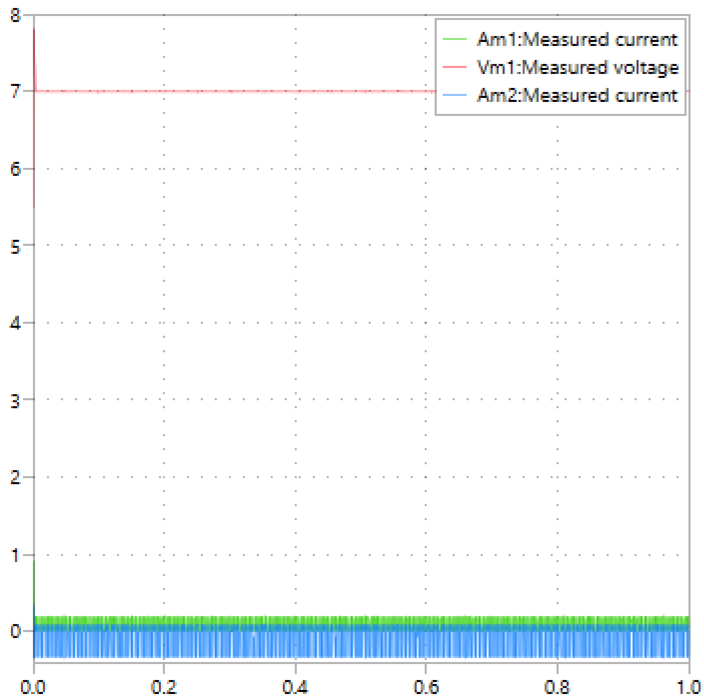

Fig 16. Output Voltage and Current Waveforms for R = 100 Ω

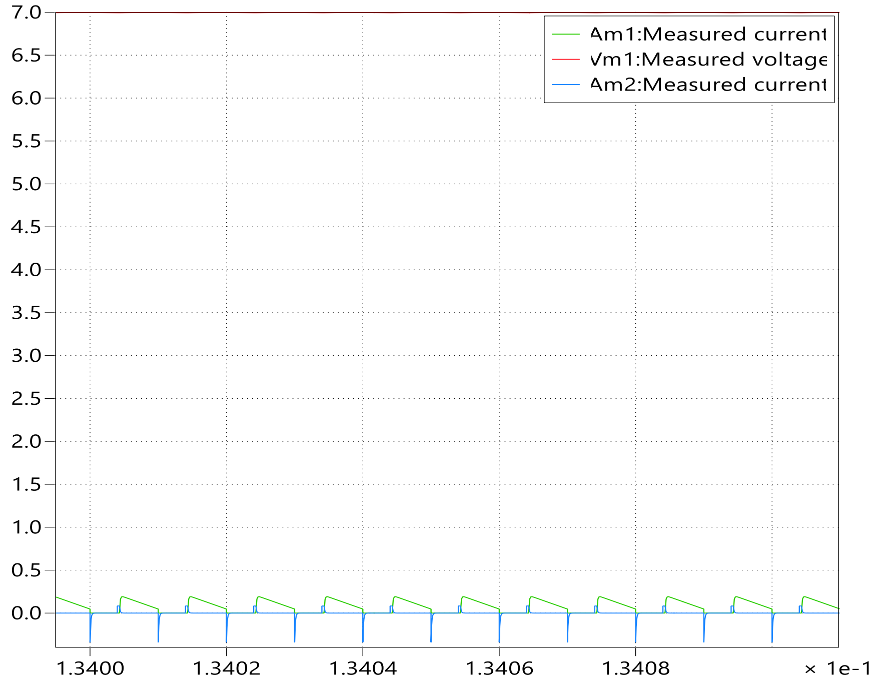

Fig17. zoomed Output Voltage and Current Waveforms for R = 100 Ω

D. Forward DC-DC Converter

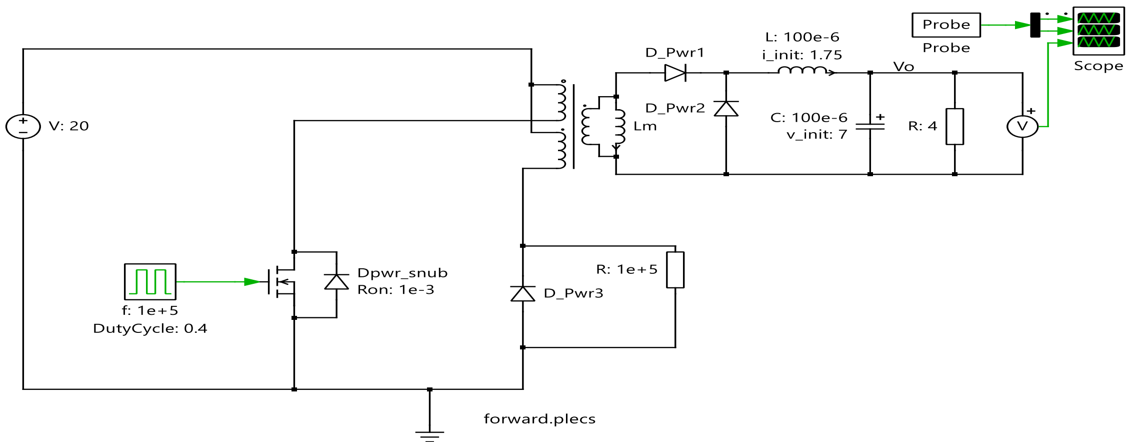

Fig 18. Forward DC-DC Converter circuit

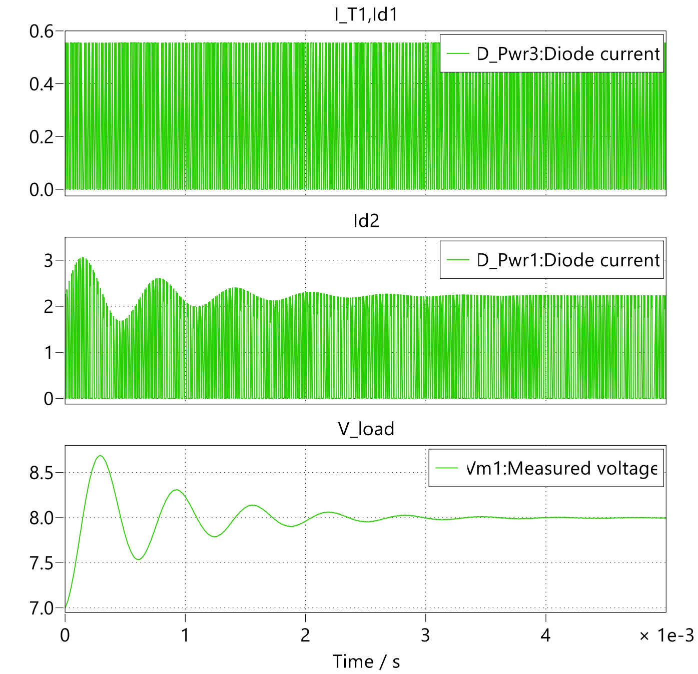

Fig 18. Simulation of Diode Current and Output Voltage in Flyback Converter

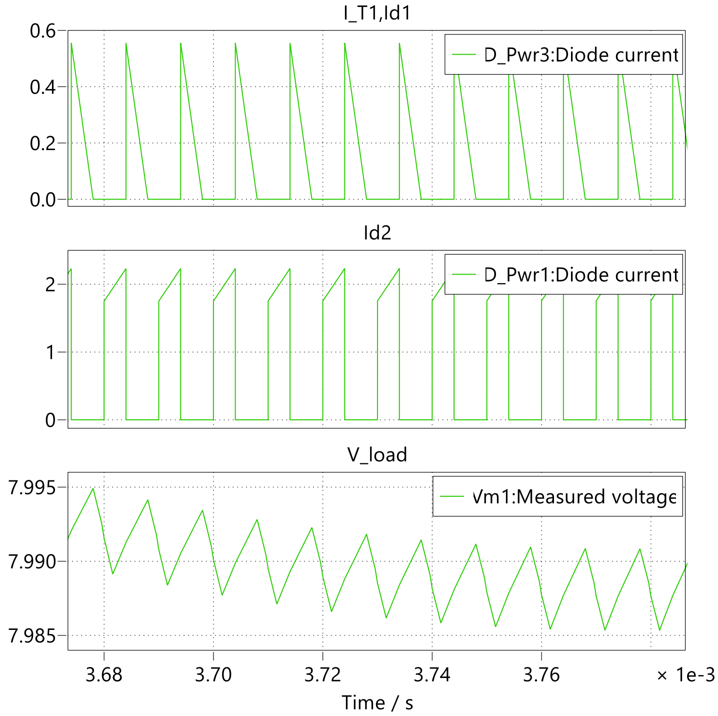

Fig 19. Zoomed Simulation of Diode Current and Output Voltage in Flyback Converter

1. *Observe the waveforms for i_1, i_3, i_{D1}, i_L and v_o. What is the relationship between i_1, i_3 and i_{D1} at the time of transitions when the switch turns-on, turns-off and the core gets demagnetized?*

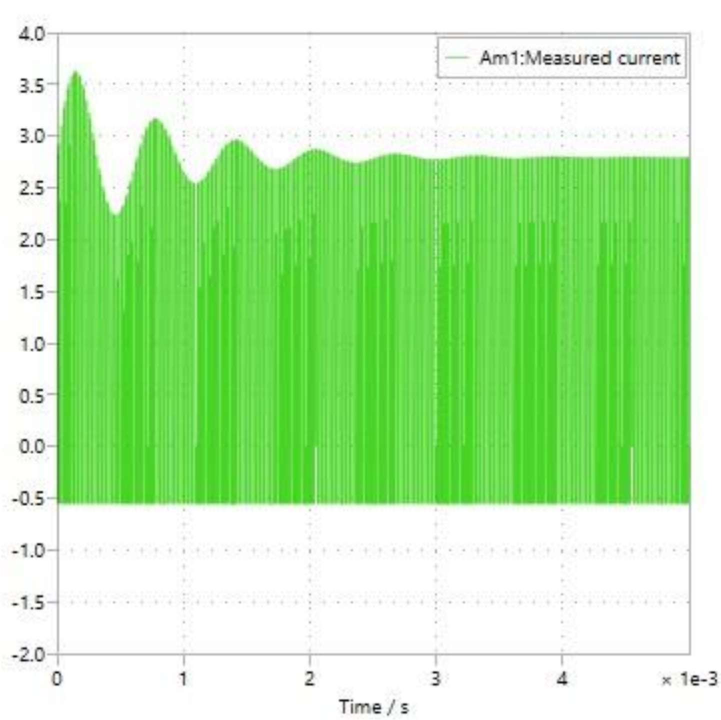

Fig 20. Inductor Current (iL) Waveform Showing Switching Transitions

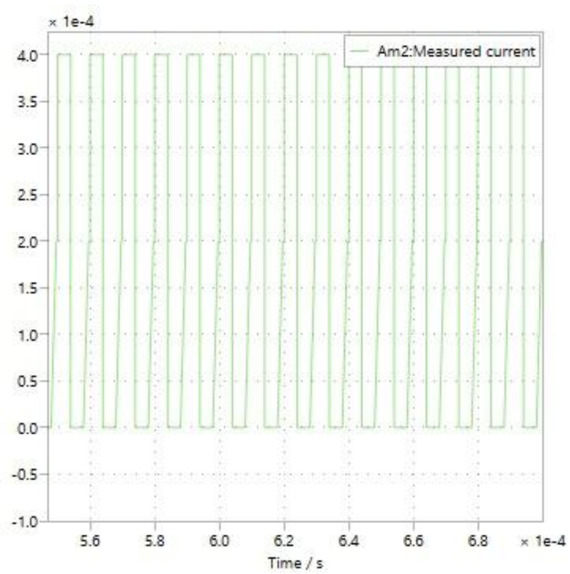

Fig.21 i3 wave- form

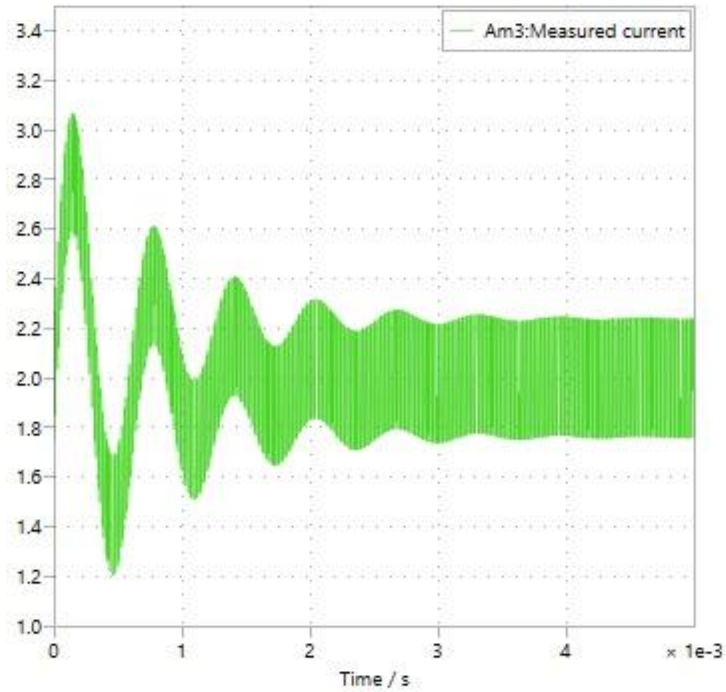

Fig.22 IL wave-form

2. *What is the voltage across the switch?*

When the MOSFET switch is turned **ON**, the voltage across it is nearly **0 V**, since the device is conducting and the drain–source terminals are effectively shorted. When the switch turns **OFF**, the reset diode becomes forward-biased during the core demagnetization phase, and the voltage across the switch rises to the sum of the input voltage and the voltage reflected from the reset winding. This relationship is expressed as:

Vsw = Vin × (1 + Np/Nr)

If the primary and reset windings have an equal number of turns (Np=Nr), the equation simplifies to:

Vsw = 2 × Vin

For the given circuit, with Vin=20 V and Np = Nr , the maximum voltage across the switch is therefore approximately **40 V**. Once the core is fully demagnetized and the reset diode stops conducting, the voltage across the switch returns close to the input voltage (≈ 20 V) and remains there until the next switching cycle starts.

3. *What is the value of R_{crit} in this converter?*

For a forward converter, the **critical load resistance ($R_{(crit)}$)** defines the boundary between continuous conduction mode (CCM) and discontinuous conduction mode (DCM). It is given by the expression

Using the given parameters L=100 μH, fs=100 kHz, and D=0.4, the critical resistance is calculated as

$$\text{Rcrit}=\frac{2\times100\times10-6\times105}{1-0.4}=\frac{20}{0.6}=\approx33.3\Omega$$

Since the load resistance R=4 Ω is much smaller than Rcrit, the converter operates well within **continuous conduction mode (CCM)**, ensuring that the inductor current never falls to zero during the switching cycle.

4. *For $R = 50\Omega$, obtain the waveforms for i_1, i_3, i_{D1}, i_L and v_o up to 0.3 ms.*

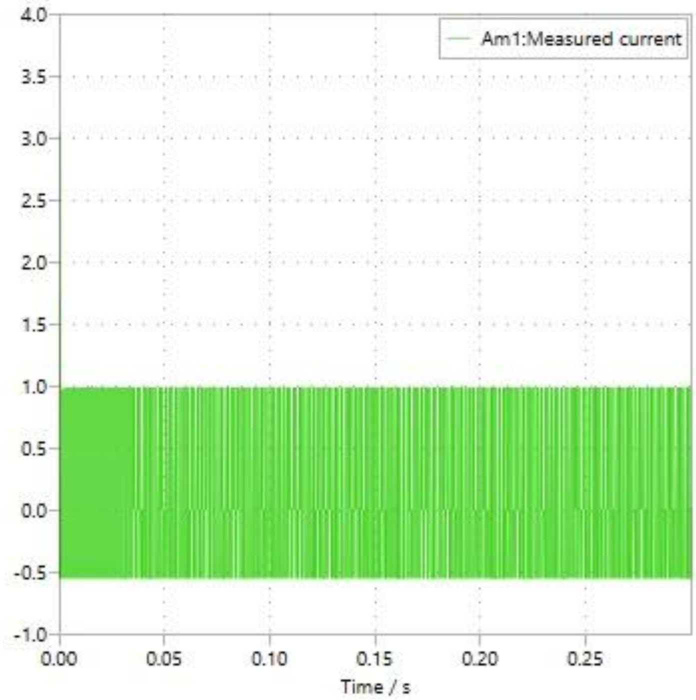

Fig.22 I1 when R=50

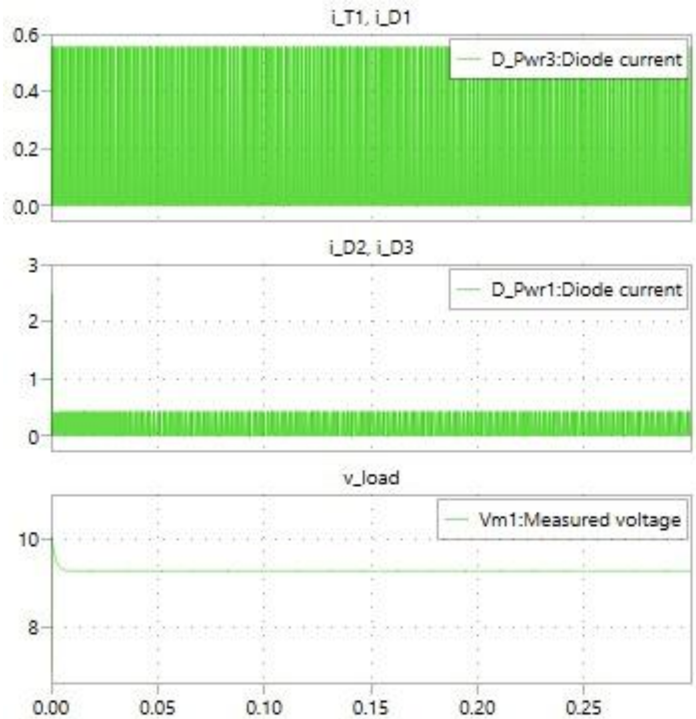

Fig.23 id1,id2,id3,vo when R=50

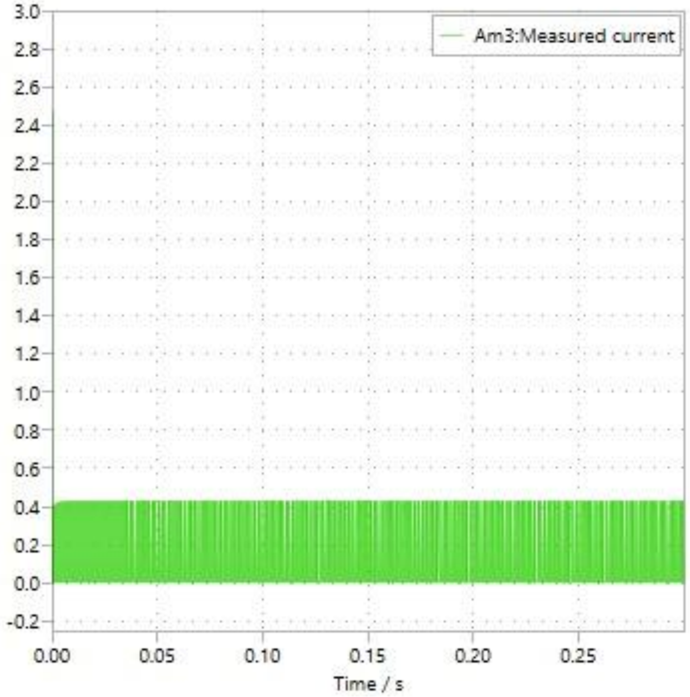

Fig.24 IL when R=50

E. Full-Bridge DC-DC converter

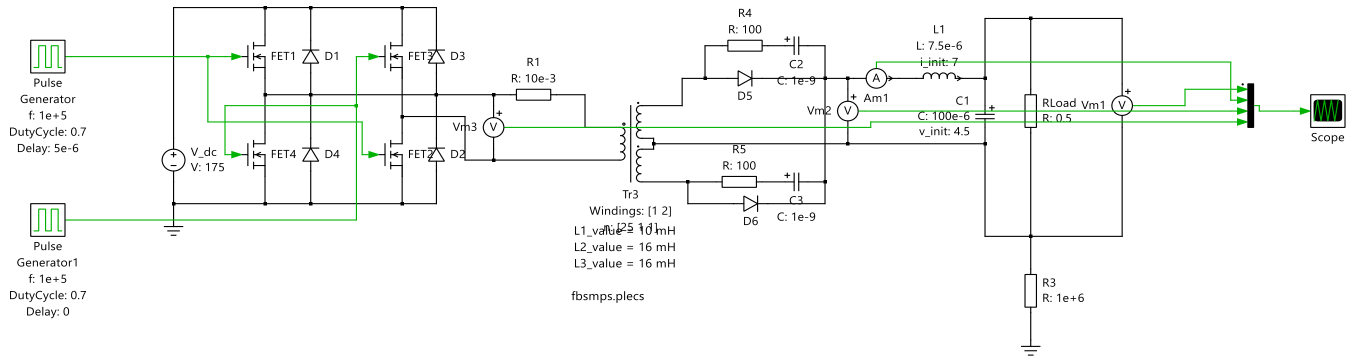

Fig 21. Full-Bridge DC-DC converter

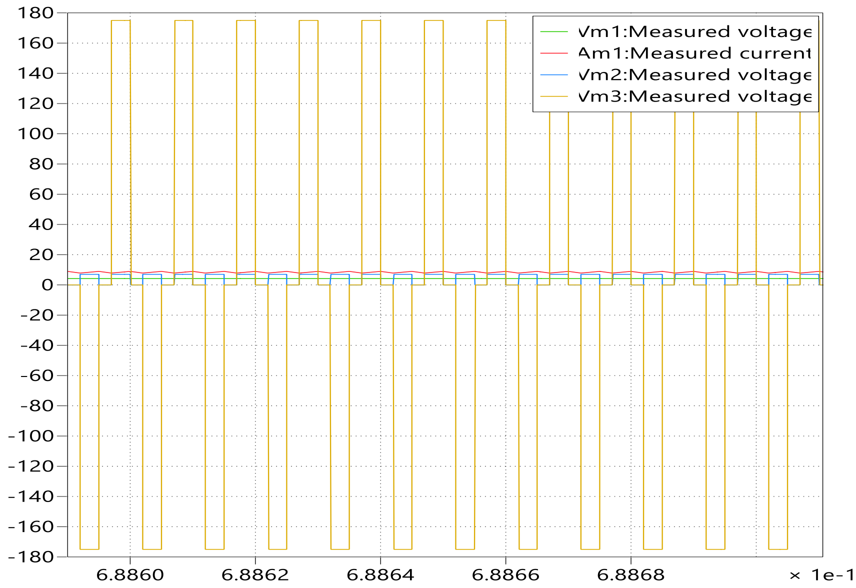

Fig 22. output waveform of Full-Bridge DC-DC converter

1. *Observe the waveforms for* v_1, v_2, v_2', v_A *and* v_o. *What are their values when all switches are off?*

When all the switches are turned **OFF**, the **primary voltage (v_1)** across the transformer becomes **0 V**, since the H-bridge is open and no drive signal is applied. Consequently, the **secondary voltages (v_2 and v_2')** are also **0 V**, as the absence of primary excitation keeps the rectifier diodes reverse-biased. The node **V_A**, located at the inductor input, is maintained at the output voltage level through the freewheeling path, resulting in **$V_A = V_o$**. The **output voltage (V_o)** remains nearly constant at its steady DC value, determined by the output capacitor **C_1** and the connected load, with only a slow drift observed — approximately **4.5 V** in this model.

Hence, under this condition: **$v_1 = v_2 = v_2' = 0$ V**, and **$V_A = V_o \approx 4.5$ V**.

5. *Plot the waveforms of i_1, i_{D1}, i_{D2} and i_L defined in Fig. 8-12. What are their values when all switches are off?*

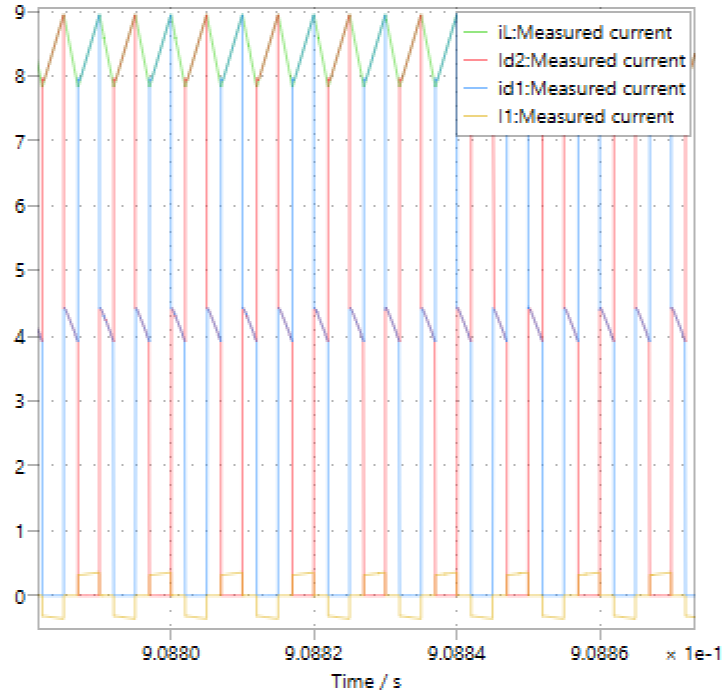

Fig 23. output waveforms of i_1, i_{D1}, i_{D2} and i_L.

Since the H-bridge is open and there is no current path through the transformer when all switches are off, the current i_1, which represents the transformer's primary current, drops to zero. The diode currents **iD_1** and **iD_2** also fall to zero during this period because there is no induced voltage on the secondary side, leaving both diodes reverse-biased. Meanwhile, the output inductor current **iL** remains nearly constant, maintaining a value of approximately **8–9 A**. This is due to the inductor's stored energy, which continues to supply the load through the freewheeling path and output capacitor, thereby sustaining continuous conduction. Hence, under this condition, **$i_1 = iD_1 = iD_2 = 0$ A**, while **$iL \approx 8$–9 A** and remains steady.

6. *Based on the converter parameters and the operating conditions, calculate the peak-peak ripple current in i_L and verify your answer with the simulation results.*

For the given parameters
$V_{in} = 175V$, $N_1 : N_2 = 1:2$; $D = 0.7$; $f_s = 100kHz$
$L = 7.5 \mu H$; $V_o = 4.5V$; $R = 0.5\Omega$,
the ripple current is calulated using

$$\Delta i_L = \frac{(V_s - V_o) D}{L \times f_s}$$

where

$$V_s = \frac{V_{in}}{(N_2/N_1)} = 87.5V$$

Substituting the values.

$$\Delta i_L = \frac{(87.5 - 4.5) 0.7}{7.5 \times 10^{-6} \times 10^5} \approx 77A.$$

$\therefore$ i_L varies between 8A and 9A
$\Delta i_L \approx 1A$

The difference occurs because the theoretical formula assumes ideal conditions, while the simulation includes real effects like transformer isolation and losses.

Theoretical ripple $\approx 77A$
Simulated ripple $\approx 1A$

This confirms that converter operates in continous conduction mode with low current ripple.

III. Conclusions

Using simulation and theoretical validation, the study effectively illustrated the performance and operation of several basic DC–DC converter topologies, including flyback, forward, buck, boost with power factor correction, and full-bridge converters. Accurate control of voltage conversion, current ripple, and steady-state operation was confirmed by the behavior of each converter, which was in line with a analytical-models.

The buck converter maintained a continuous conduction mode while efficiently stepping down the input voltage with little ripple. The PFC circuit produced a regulated output voltage with a power factor close to unity and enhanced the quality of the input current. While the forward converter guaranteed effective energy delivery and total magnetic core reset, the flyback converter demonstrated appropriate transformer isolation and sequential energy transfer between primary and secondary windings. The full-bridge converter showed steady, low-ripple output appropriate for high-power applications and balanced operation.

The accuracy of the applied design equations and the converters' proper operation under designated duty cycles and switching frequencies were confirmed by the close alignment of simulated and theoretical results across all experiments. Together, these experiments improved knowledge of the fundamentals of switching power conversion and highlighted how crucial converter design, control, and analysis are to contemporary power electronics systems.

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Thank you